MATH 3020 Syllabus

Course Registration

- Math 3020 Applied Probability and Statistics for Computer Science
- CRN: 86019
- Fall semester, 2026

Instructor Contact Information

- Instructor: Dr. Xiaojing Ye
- Office Hours (Virtual): Tuesdays and Thursdays, 10:15 AM - 11:30 AM
- Office Hour Zoom Meeting ID / Passcode: See iCollege course page.

iCollege

- Course related materials are posted on iCollege at https://icollege.gsu.edu/

Course Meeting

- Format: Virtual on Zoom
- Class Time: Tuesdays and Thursdays, 12:45 PM - 2:00 PM
- Class Zoom Meeting ID / Passcode: See iCollege course page.

Prerequisite

- Grade of C or higher in MATH 2212 (Calculus II)

Required Textbook

- Probability and Statistics for Computer Scientists, 3rd edition, Michael Baron (2019). CRC Press.

Course Content

Applied Probability and Statistics for Computer Science. Prerequisite: MATH 2212 with a C or higher. This course covers theory and applications of probability models, random variables, discrete and continuous probability distributions, joint and conditional distributions, sampling distributions, central limit theorem, hypothesis testing, confidence intervals, and exposure to simple linear regression. Time-to-failure probability models are considered.

Learning Outcomes

Upon successful completion of the course, students should be able to:

1. Describe sample spaces and events, perform set operations on events, and apply the axioms and basic rules of probability.

2. Compute probabilities using equally likely outcomes, permutations, combinations, conditional probability, independence, Bayes Rule, and the Law of Total Probability.

3. Define discrete random variables and random vectors and determine probability distributions, joint distributions, marginal distributions, and independence.

4. Compute and interpret expectation, variance, standard deviation, covariance, and correlation, and apply Chebyshev’s inequality.

5. Recognize and apply common discrete probability distributions, including Bernoulli, Binomial, Geometric, Negative Binomial, and Poisson distributions, as well as the Poisson approximation to the Binomial distribution.

6. Work with continuous random variables and probability densities, including joint and marginal densities, expectation, and variance.

7. Recognize and apply Uniform, Exponential, Gamma, and Normal distributions, including applications involving waiting times and time-to-failure models.

8. Apply the Central Limit Theorem and use the Normal distribution to approximate Binomial probabilities.

9. Distinguish populations from samples and parameters from statistics, and compute descriptive measures including means, medians, quantiles, percentiles, quartiles, variances, standard deviations, standard errors, and interquartile ranges.

10. Construct and interpret graphical summaries of data, including histograms, stem-and-leaf plots, boxplots, scatter plots, and time plots.

11. Estimate unknown population parameters using the method of moments and maximum likelihood and estimate standard errors.

12. Construct and interpret confidence intervals for population means, differences between means, proportions, population variances, and ratios of variances, and determine appropriate sample sizes for specified precision.

13. Formulate and conduct hypothesis tests, distinguish Type I and Type II errors, determine rejection regions and power, and perform Z-tests and T-tests for means and proportions.

14. Compute and interpret P-values and explain the relationship between two-sided hypothesis tests and confidence intervals.

15. Perform inference for variances using the Chi-square distribution and compare two population variances using the F-distribution.

Textbook Sections Covered

- Chapter 2 Probability
    - 2.1 Events and their probabilities
    - 2.2 Rules of Probability
    - 2.3 Combinatorics
    - 2.4 Conditional probability and independence
- Chapter 3 Discrete Random Variables and Their Distributions
    - 3.1 Distribution of a random variable
    - 3.2 Distribution of a random vector
    - 3.3 Expectation and variance
    - 3.4 Families of discrete distributions
- Chapter 4 Continuous Distributions

  - 4.1 Probability density
  - 4.2 Families of continuous distributions
  - 4.3 Central Limit Theorem
- Chapter 8 Introduction to Statistics
  - 8.1 Population and sample, parameters and statistics
  - 8.2 Descriptive statistics
  - 8.3 Graphical statistics
- Chapter 9 Statistical Inference I
  - 9.1 Parameter estimation
  - 9.2 Confidence intervals
  - 9.3 Unknown standard deviation
  - 9.4 Hypothesis testing
  - 9.5 Inference about variances (if time permits)

Course Evaluation

Quizzes, tests, and final are closed-book and closed-notes.

- Quiz (30%)
  - 8 quizzes (lowest 2 will be dropped). **No make-up**.
  - Virtual in iCollege. LockDown browser and webcam required.
- Test (40%)
  - 3 tests (lowest 1 will be dropped).
  - Virtual in iCollege. LockDown browser and webcam required.
  - Held at **selected 3 lecture class periods**. See dates in iCollege course webpage.
- Final Exam (30%)

    - Date/time: **12/15 10:45 AM - 1:15 PM**.
    - Location: TBD. (Per university policy, final exams of online courses must be in-person.)

Remarks on iCollege Quizzes and Tests

- Request for make-up test or final with verifiable excuses must be submitted for approval **at least 2 weeks before** the scheduled day (see above)
- All virtual quizzes and tests are in “Assessment->Quizzes” in iCollege.
- Computer with Windows or macOS only. No iPad.
- Scientific calculator is provided by LockDown Browser. Using any other technology could result in an academic dishonesty referral.
- Requires showing ID (Panther ID or Driver’s License) before start.

Grading Scale

- **A+**: 97%–100% | **A**: 93%–96% | **A-**: 90%–92%
- **B+**: 87%–89% | **B**: 83%–86% | **B-**: 80%–82%
- **C+**: 77%–79% | **C**: 70%–76%
- **D**: 60%–69%
- **F**: Below 60%

Other Faculty Initiated Withdrawals

If you stop attending the course before the semester midpoint (October 13, 2026), you may be administratively withdrawn from the course and receive a withdrawal grade. Attending the course implies consistent class attendance and active involvement on iCollege.

Failure to meet the following requirements will be considered as lack of attendance:

- If you do not attend any of the class periods during the first week of the semester, you will be administratively withdrawn from the course as “Never Attended”.

- If you do not access any class resources on iCollege, like never open the quiz on iCollege, you will be administratively withdrawn from the course as “Never Attended”.

Excused Absences and Makeup Policy

- GSU has a process for students seeking excused absences through the Dean of Students (DOS) Office. In case of any emergency, students must promptly submit documentation to https://deanofstudents.gsu.edu/student-assistance/professor-absence-notification/. Professors will then be notified by the DOS of any excused absences without the need to manage medical or other pertinent information individually. Making up any graded assignments missed during this period of absence is allowed only when we receive such a notification from the DOS Office.

- All other, i.e. non-medical, special circumstances should be communicated directly and promptly to the instructor, together with proper verifiable documentation (e.g., a doctor’s note or a police report is necessary). The decision of allowing or not making up missed work in these cases will be at the instructor’s discretion.

- If a test grade is zero due to violation of academic integrity (see the respective policy below), then it will stand as 0 (zero) and will not be replaced.

Calculator Policy

For all graded assignments you are allowed to use a non-programmable, non-graphing, non-symbolic-solving scientific calculator. Use of mobile phones is not permitted in place of a calculator under any circumstances. If you are in any doubt as to whether your existing or planned calculator is suitable, you must get it passed by your instructor prior to its use. Breaking these rules will be treated as cheating according to the university guidelines below.

Writing in Mathematics

Your grades on all written assignments will be based on clear presentation as well as correct mathematics. It is often a good idea to model your writing on the examples worked in class or posted online. If you are unsure of whether written work is acceptable, you should ask about it. Please be aware that the process, and not merely the final answer, is critical to your understanding of the material and your success in the course. Precise, effective writing will be rewarded. Careless or incomplete work will be penalized, even if by chance it leads to a correct numerical answer. This means your final answer is worth less than the procedure you used to get your answer.

Accommodations

Students who wish to request any accommodations for a disability may do so by registering with the Access and Accommodations Center (https://access.gsu.edu/). Students may only be accommodated upon issuance by the Center of a signed Accommodation Plan and are responsible for providing a copy of that plan to instructors of all classes in which accommodations are sought.

Academic Integrity Policy

Cheating/plagiarism will not be tolerated on any work. Academic dishonesty on the final exam may result in an F for the course. A first occurrence will result in a grade of 0 on the assignment for all concerned parties as well as an Academic Dishonesty form being filed with the Dean's Office. A second occurrence will result in a grade of F for the course for the concerned parties and a second Academic Dishonesty form being filed. (See also the University's policy on Academic Honesty at http://codeofconduct.gsu.edu/.)

Copyright and Honesty Clause

All content created in this course, including videos, handouts, etc., may be used only by students enrolled in the course for purposes relating to the course. The selling, sharing, publishing, presenting, or distributing of instructor-prepared course lecture notes, videos, audio recordings, or any other instructor-produced materials from any course for any commercial or non-commercial purpose is strictly prohibited, unless explicit written permission is granted in advance by the course instructor. This includes posting any materials on websites such as Chegg, Course Hero, OneClass, Stuvia, StuDocu, and other similar sites. Unauthorized sale or commercial or non-commercial distribution of such material is a violation of the instructor's intellectual property and the privacy rights of students attending the class and is prohibited. Failure to abide by these limitations constitutes a violation of the Policy on Academic Honesty and will be treated accordingly.

Students Conduct Guidelines

Appropriate conduct is expected from all students. Arrive on time (i.e. join the class lecture meeting on time), and do not leave the meeting early. If you must leave early for some reason, please inform me in the chat prior to class. Please always mute yourself and open your camera while in the meeting room. Text messaging, instant messaging, emailing, etc. during class is strictly prohibited and is grounds for dismissal. If you are using your cell phone or computer or another device for tasks that are not math related, or talking, or otherwise disrupting students, you will be asked to leave. After the third incident you will be administratively removed from the class (as per the Student Handbook).

Withdrawal Policy

- Undergraduates: If you withdraw from this class on or before the Midpoint of the semester (5:00pm on Tuesday, October 13, 2026), you will receive a WP regardless of your performance. The computer will then turn this into a W or a WF depending on how many cumulative withdrawals you have in the University. Voluntary withdrawals after the Midpoint are not allowed.

- Others: If you withdraw from this class on or before the Midpoint of the semester (5:00pm on Tuesday, October 13, 2026), you will receive a W or a WF depending on your performance. You must be passing (70 average or better at the time of withdrawal) to receive a W.

Additional Information

- Ask course-related questions in office hours.

- Visit the STEM Tutoring Center, located in the GSU Sports Arena, room 110 (https://cas.gsu.edu/academics-admissions/undergraduate-learning/stem-education-programs/stem-tutoring/).

- Visit the Counseling and Testing Center: learning assistance, test anxiety classes, student support services (http://counselingcenter.gsu.edu/, 404-413-1640).

- Student Feedback Your constructive assessment of this course plays an indispensable role in shaping education at Georgia State University. Upon completing the course, please take time to fill out the online course evaluation.

THIS SYLLABUS PROVIDES A GENERAL PLAN FOR THE COURSE; DEVIATIONS MAY BE NECESSARY.